

# YOLO Training Report (Stage 1)

Run directory: ..\Runs\Stage 1

## Configuration

| Parameter    | Value                |
|--------------|----------------------|
| model        | ../Models/yolo26s.pt |
| epochs       | 50                   |
| imgsz        | 1024                 |
| batch        | 8                    |
| optimizer    | AdamW                |
| lr0          | 0.001                |
| lrf          | 0.001                |
| patience     | 10                   |
| device       | 0                    |
| workers      | 4                    |
| auto_augment | None                 |

## Training Summary

| Metric           | Value  |
|------------------|--------|
| Total epochs run | 51     |
| Best epoch       | 25     |
| mAP50 (best)     | 0.9950 |
| mAP50-95 (best)  | 0.7125 |
| Precision (best) | 0.9972 |
| Recall (best)    | 0.9926 |

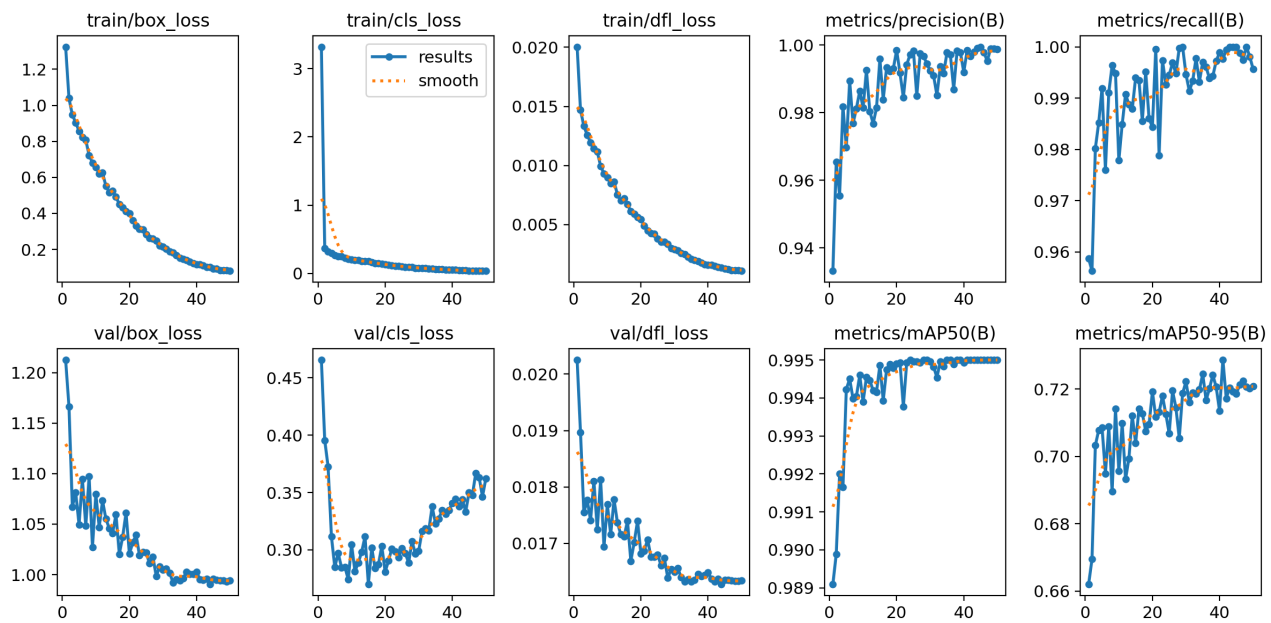


Figure: loss and metric curves over training epochs.

# Per-Class Performance

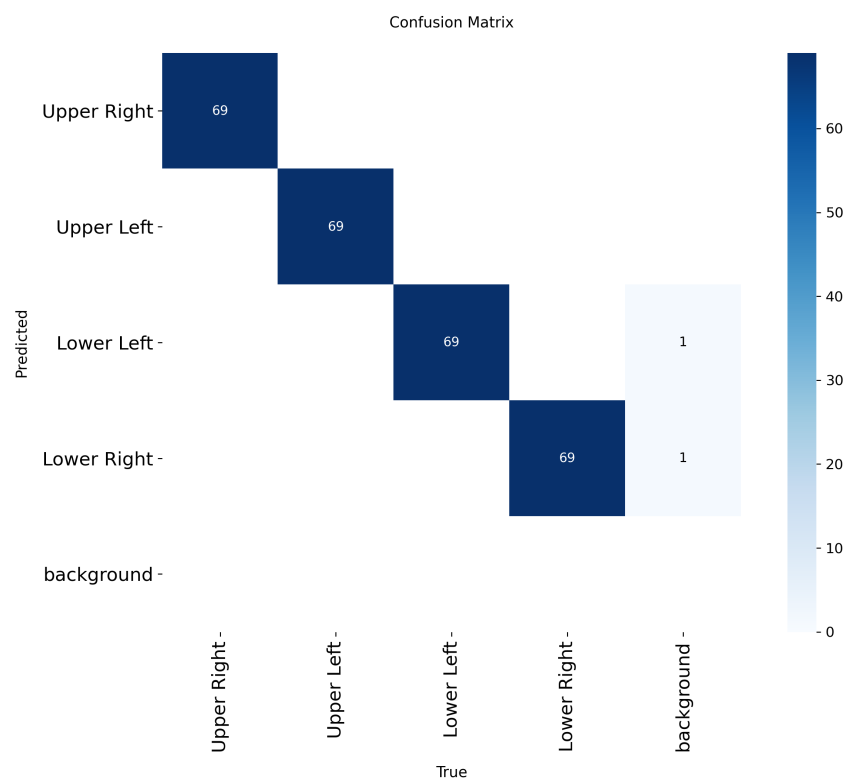
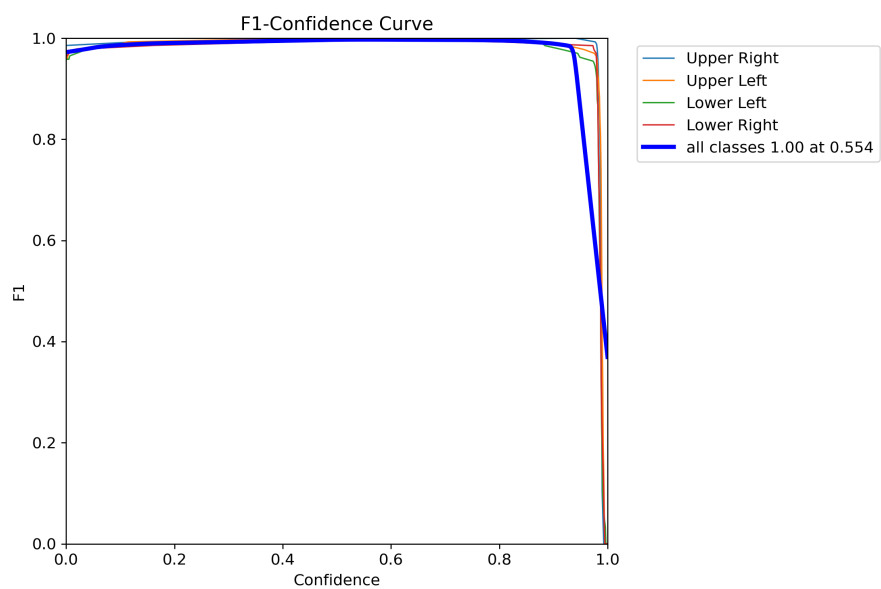
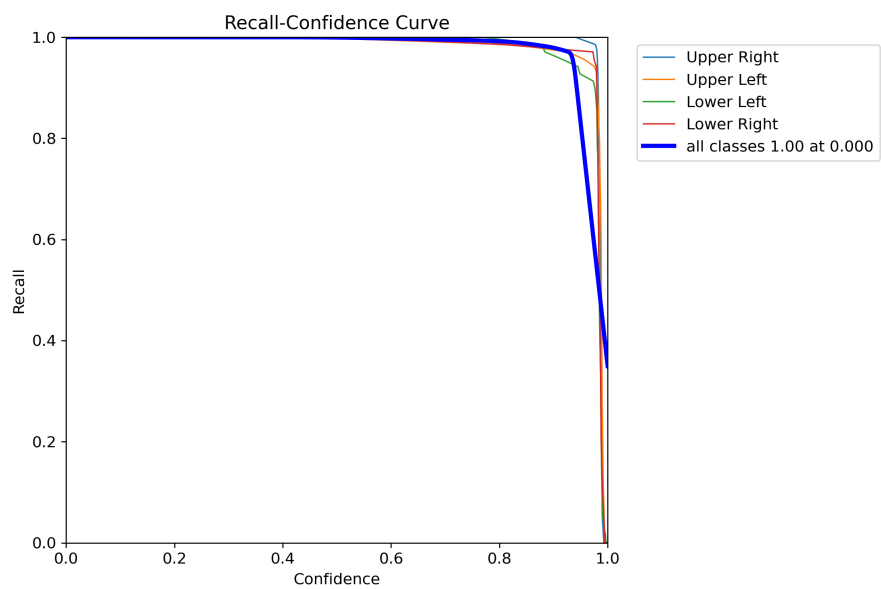
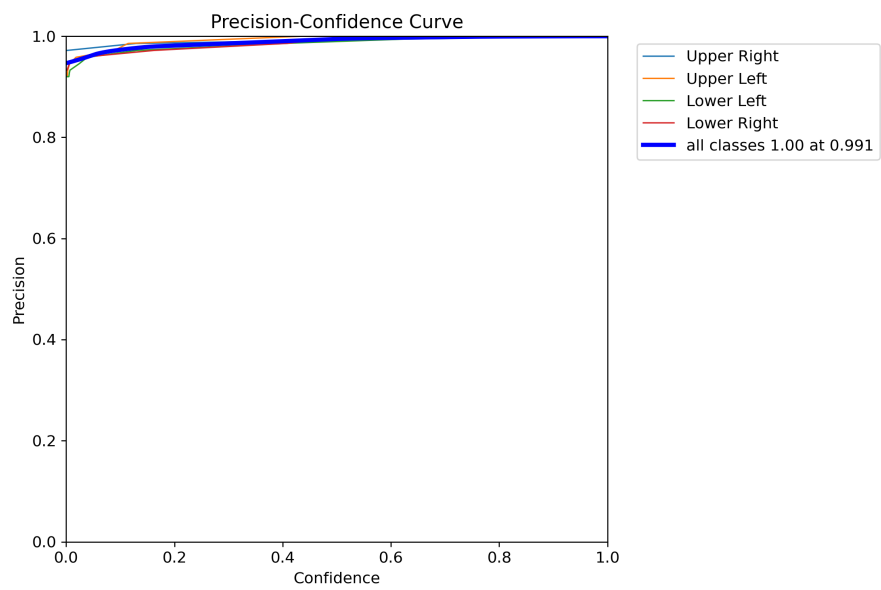
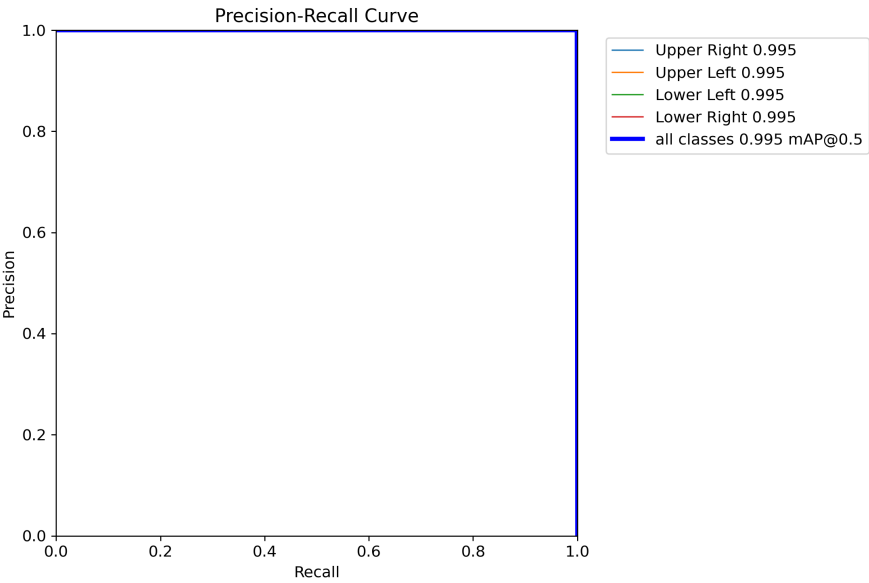


Figure: confusion matrix on the validation set.

# Precision / Recall / F1 Curves

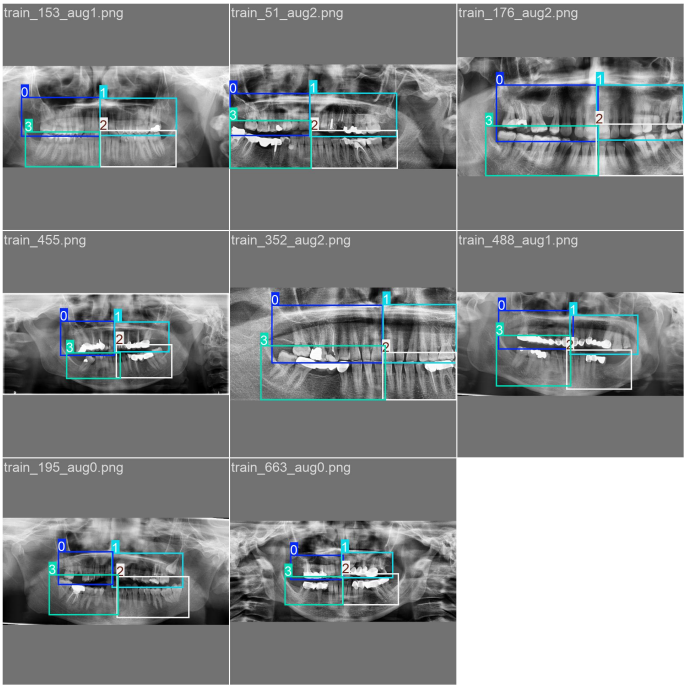


BoxF1\_curve.png

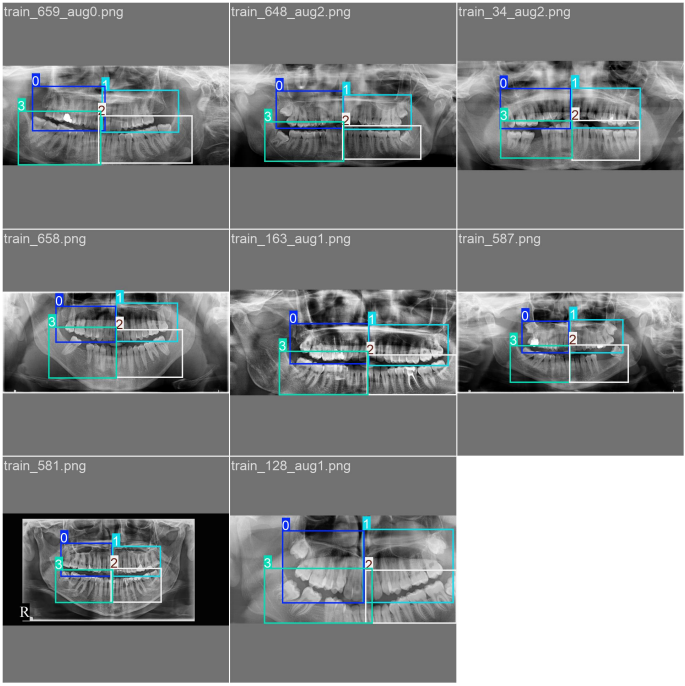


BoxPR\_curve.png

# Sample Training Batches

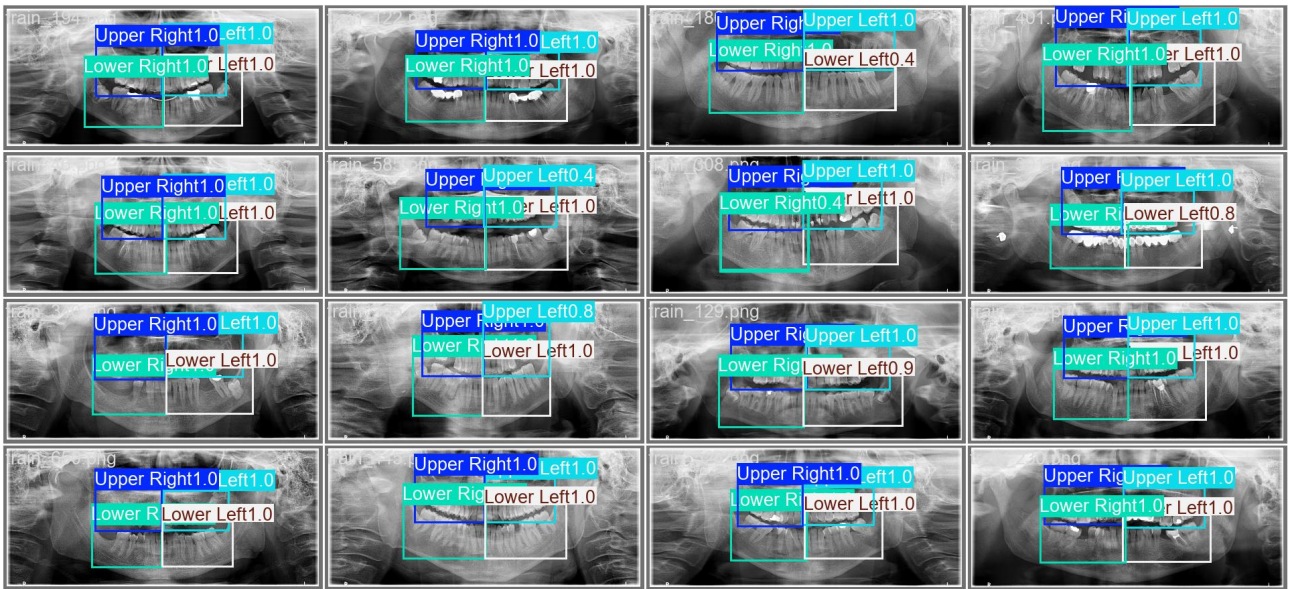


train\_batch0.jpg

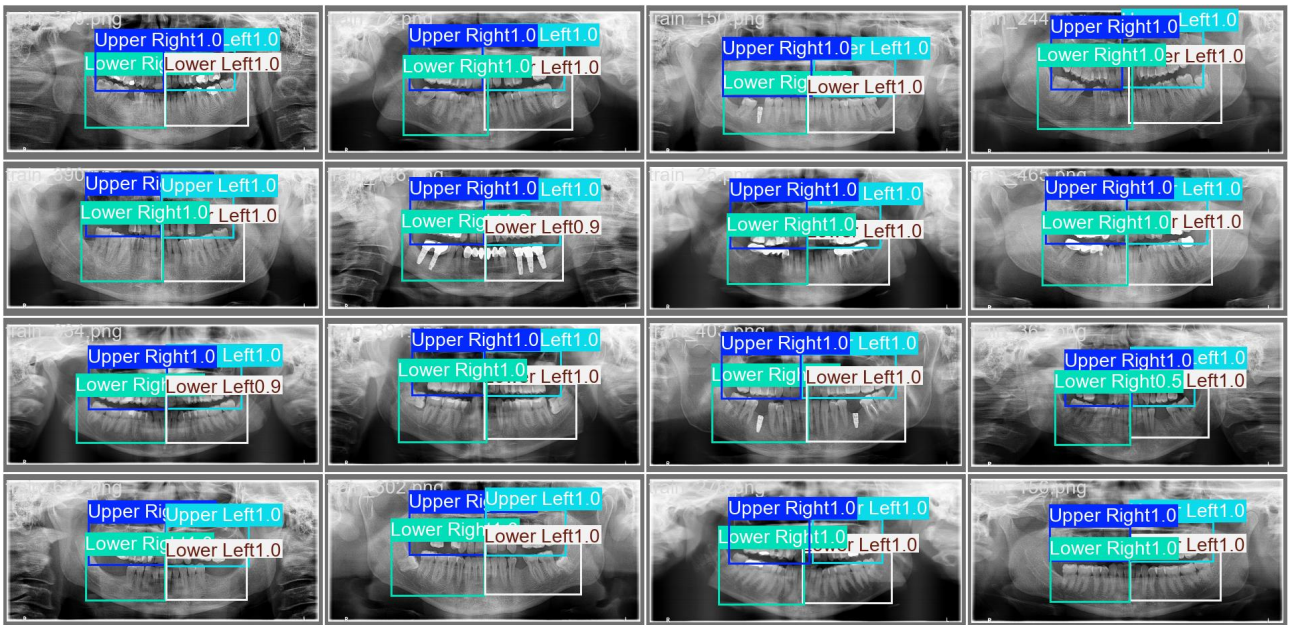


train\_batch1.jpg

# Sample Validation Batches



val\_batch0\_pred.jpg

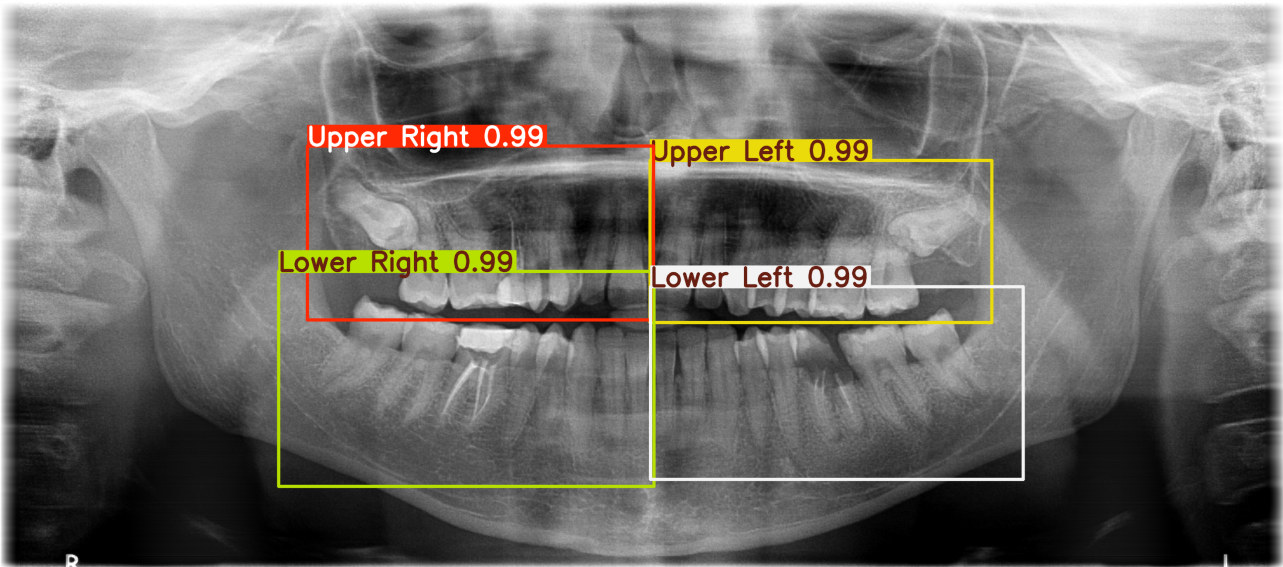


val\_batch1\_pred.jpg

## Test Set Predictions

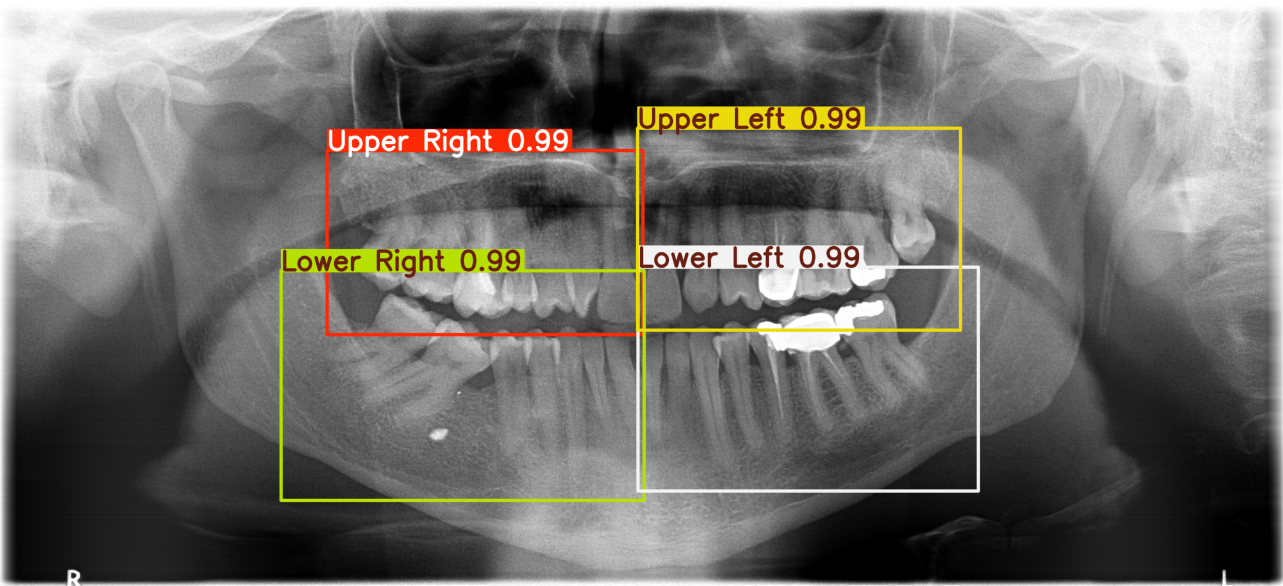
Qualitative predictions on held-out images not used in train/val (disease-only annotated images kept aside for visual inspection).

Image: train\_2



Full pipeline output

Image: train\_80



Full pipeline output

Image: train\_241

Quadrant-level predictions:



Lower Left



Upper Right

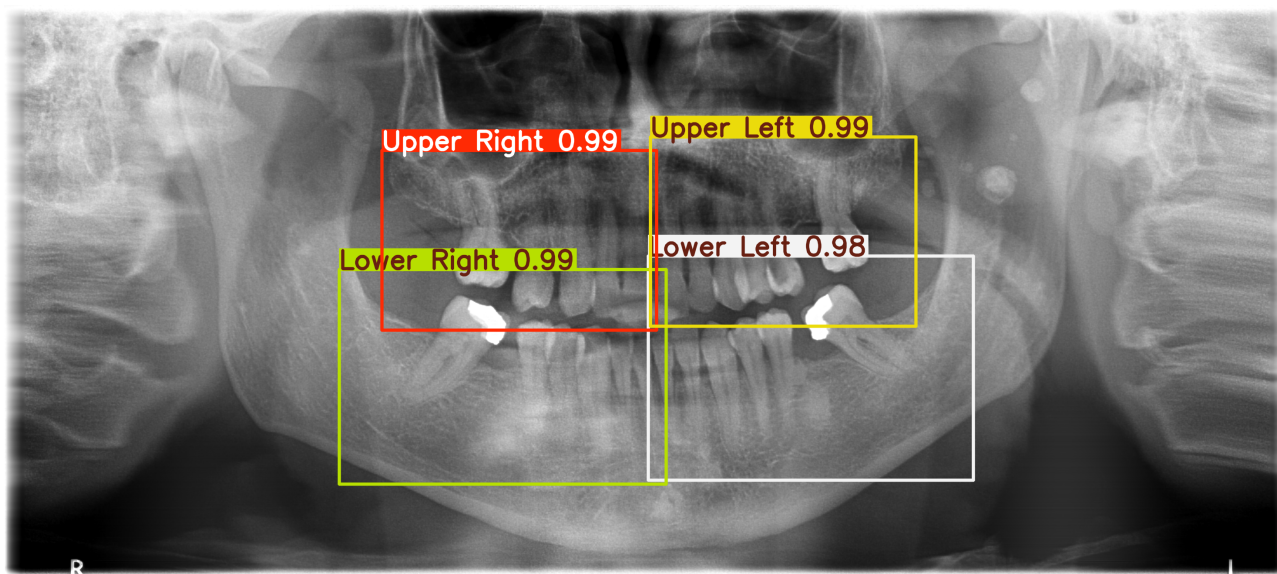


Upper Left



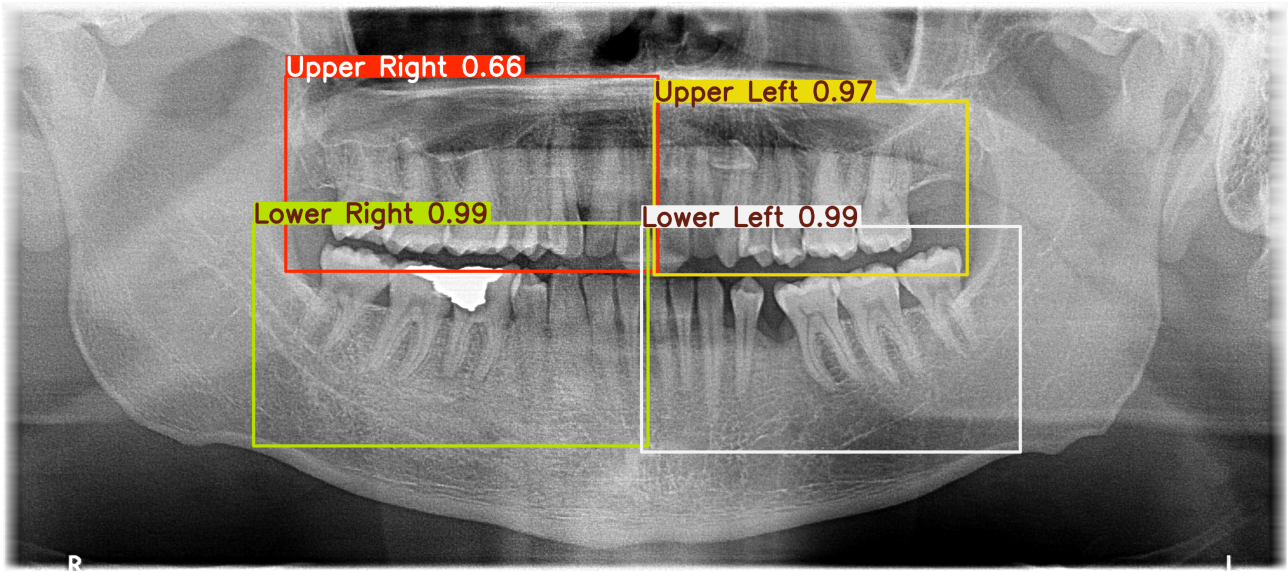
Lower Right

Image: train\_284



Full pipeline output

Image: train\_344



Full pipeline output

## Model Artifacts

| File    | Exists | Size (MB) |
|---------|--------|-----------|
| best.pt | Yes    | 19.4      |
| last.pt | Yes    | 19.4      |